TECHNICAL REPORT: AIRFLOW SIMULATION

Project : Batiment_R+4

Client : Promoteur_X

Location : Lyon_69

1. NETWORK SUMMARY

Category	Parameter	Value
Air properties	Temperature used (°C)	20.00
	Altitude (m)	170.00
	Density (kg/m³)	1.180
	Dynamic viscosity (Pa·s)	1.813e-05
General metrics	Ventilation type	Supply
	Design volumetric flow rate (m³/h)	2200.00
Fan: fan_01	Fan label	Main_Fan
	Description	Fresh air supply
	Nominal target efficiency (%)	75%
	Fan flow (m³/h)	2200.00
	Critical node	of_04
	Cumulative static (Pa)	59.84
	Exit dynamic (Pa)	19.57
	Total pressure (Pa)	79.40
	Shaft input power (W)	64.70

Operating cost projections	Total shaft input power (W)	64.70
	Daily operating cycle (h/day)	10.00
	Annual operating cycle (days/year)	250.00
	Annual energy usage (kWh)	161.75
	Electricity Unit Price (€/kWh)	0.25
	Estimated annual opex (€)	40.44

2. TERMINAL BALANCING ANALYSIS

Terminal Node	Static (Pa)	Dynamic (Pa)	Total (Pa)	Balance (Pa)	K-Factor
of_01	44.84	19.57	64.41	14.99	0.77
of_02	54.44	18.47	72.92	6.48	0.35
of_03	39.03	20.62	59.64	19.76	0.96
of_04	59.84	19.57	79.40	0.00	0.00

3. DUCT PERFORMANCE ANALYSIS

Duct Name	Flow Rate (m³/h)	Velocity (m/s)	Linear Loss (Pa)	Lin. Loss/m (Pa/m)
main_trunk	2200.00	3.84	0.75	0.373
to_junction_A	2200.00	4.86	2.01	0.669
office_1	550.00	4.49	2.72	1.359
office_2	500.00	4.34	2.66	1.330
transit_to_B	1150.00	4.52	2.49	0.830
office_3	600.00	4.63	2.76	1.381
office_4	550.00	4.49	2.72	1.359

4. HYDRAULIC AUDIT DETAILS

Duct Name	Reynolds	Friction Lambda	Sing. Loss (Pa)	Zeta Total
main_trunk	112523	0.0193	0.00	0.000
to_junction_A	126588	0.0192	0.43	0.031
office_1	53741	0.0210	38.94	3.269
office_2	50213	0.0213	48.60	4.373
transit_to_B	88228	0.0207	1.53	0.127
office_3	57084	0.0207	29.07	2.298
office_4	53741	0.0210	49.92	4.190

5. ACOUSTIC NOISE RISK ASSESSMENT

Duct Name	Topology Type	Velocity (m/s)	Noise Lw [dB(A)]	Acoustic Status
main_trunk	Main Trunk	3.84	31.20	OPTIMAL
to_junction_A	Intermediary	4.86	35.30	OPTIMAL
office_1	Terminal Runout	4.49	27.90	OPTIMAL
office_2	Terminal Runout	4.34	26.90	OPTIMAL
transit_to_B	Intermediary	4.52	31.20	OPTIMAL
office_3	Terminal Runout	4.63	28.80	OPTIMAL
office_4	Terminal Runout	4.49	27.90	OPTIMAL

6. ACOUSTIC SIZING RECOMMENDATIONS (CRITICAL)

Duct Name	Status	Vitesse (m/s)	Circ. Diam (mm)	Rect. WxH (mm)
No critical ducts identified				

7. SOLVER METADATA

Performance Metric	Result
Execution time (s)	0.119
Convergence status	stable
Residual error (m³/s)	8.2e-10
Tolerance targeted (m³/s)	1e-09
Iterations performed	477
Max iterations allowed	1000000
Timestamp	26/05/2026 19:41:13

8. NETWORK DIAGRAM